

Statistical Analysis of Car Prices and Attributes

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Introduction

This report presents a statistical and visual analysis of a car dataset containing information about multiple car brands and models, engine sizes, fuel types, years of manufacture, mileage, and prices in Great British Pound. The primary goal is to explore how mileage, engine size, and manufacturer characteristics influence car prices. The analysis combines descriptive statistics with three major plot types: relational (scatter plot), categorical (bar chart), and statistical (heatmap). Each plot reveals unique insights about patterns and relationships between the attributes of cars in the dataset.

The dataset was preprocessed by removing missing values and converting relevant columns to numeric types. Four main statistical moments (mean, standard deviation, skewness, and kurtosis) were computed for the Price (GBP) column to assess its distribution and variability.

1. Relational Analysis: Mileage vs Price

The scatter plot (see **Figure 1**) illustrates the relationship between car mileage and price, with each dot representing a car, while the colour of the dot indicates the **fuel type** (Petrol, Diesel, or Hybrid). There is a clear negative correlation between mileage and price, meaning that cars with higher mileage tend to have lower resale value. This is expected because higher mileage reflects greater wear and depreciation. The relationship also varies across fuel types: Diesel cars appear largely concentrated within the mid-range for both mileage and price. This suggests they hold their value fairly well, but they are not always more expensive than petrol cars. Hybrid vehicles tend to form smaller, distinct clusters, often commanding higher prices despite lower mileage levels, reflecting modern consumer preferences and sustainability trends.

Summary of Price Statistics:

Mean Price = £13,828.90

Standard Deviation= £16,416.52

Skewness = 2.85 (right-skewed)

Kurtosis = 12.81 (leptokurtic)

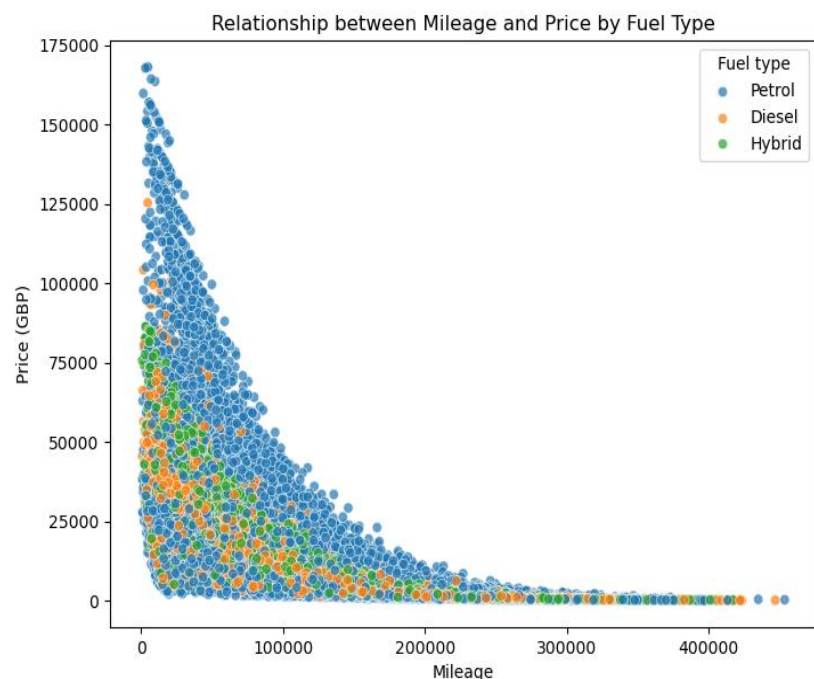


Figure 1: Scatter plot of Mileage vs Price by Fuel Type

2. Categorical Analysis: Average Price by Manufacturer

The bar chart (see **Figure 2**) compares the average prices of cars from the top five manufacturers in the dataset. Porsche and BMW clearly stand out with the highest average prices, reflecting their positioning as luxury and premium brands. While, Toyota, Ford, and Volkswagen (VW) show much lower average prices, indicating their focus on affordable, mass-market cars. In general, the chart highlights a clear pricing hierarchy, suggesting that

manufacturer reputation and brand positioning strongly influence resale value beyond purely mechanical characteristics like mileage or engine size.

3. Statistical Analysis:

Correlation Heatmap

The correlation heatmap (see **Figure 3**) reveals the relationships among numerical attributes: engine size, year of manufacture, mileage, and price. Key findings include;

Mileage and Price (-0.63): strong negative correlation, this supports the earlier scatter plot findings that higher mileage reduces resale value.

Year of Manufacture and Price (0.71): positive correlation, newer cars are typically more expensive. indicating that vehicle age is a major factor influencing price.

Engine Size and Price (0.40): moderate positive correlation, larger engines often command higher prices, though not always in a perfectly proportional way.

Price is most strongly influenced by vehicle age (year of manufacture) and mileage, with newer and less-driven cars commanding higher prices. Engine size also affects price but not as strongly as age or mileage.

4. Conclusion:

The analysis confirms key economic principles in the automobile market:

- Cars with **larger engines** and **newer manufacture years** attract higher prices.
- **Mileage** remains a critical depreciation factor, inversely related to price.
- **Fuel type** affects value perception; electric and

hybrid cars command higher prices.

In summary, this analysis reveals that car pricing is shaped by a combination of condition (mileage and age), performance attributes (engine size), and brand market segment. These insights can help buyers, sellers, and dealerships to better understand the value trends within the used-car market.

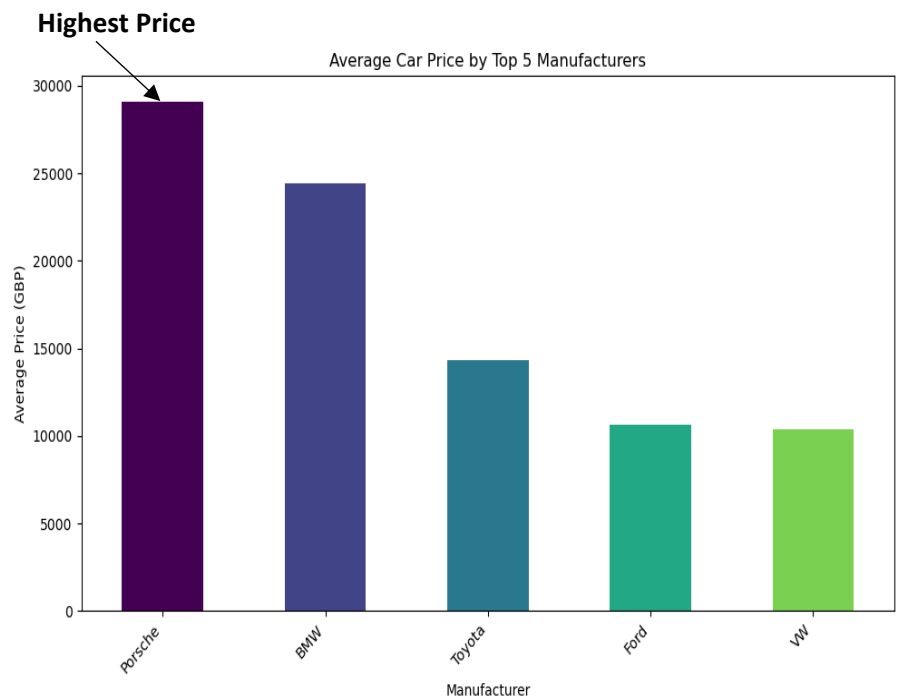


Figure 2: Bar chart of Average Price by Top 5 Manufacturers

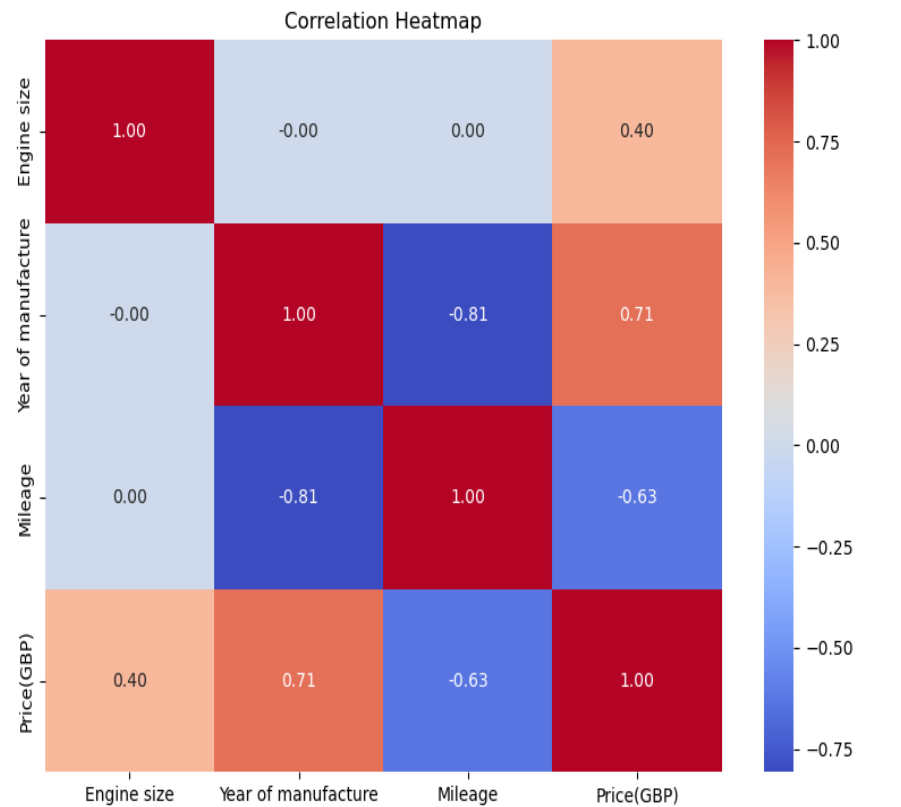


Figure 3: Correlation Heatmap of Key Attributes